The audio analysis — in plain English

What it does, what already works, and what it will let us do. The detailed maths is in

identifiability-of-room-modes.md; the programme plan is AirMatrix's

`PLANS/PLAN-airmatrix-audio-lab.md`; the code is `src/pacland/spectral/` here and `rob-meter` / `rob-dsp`

in the AirMatrix repo.

In one sentence

It turns a PA system into an **audio lab that can certify what it measured** — and, before any of that, it

answers the question every room-correction product skips: *how much of a room can this microphone setup

actually know?*

What already works — measured, not planned

These are in the tree and verified with figures, not promises:

capability	verified
loudness (K-weighted, M/S/I, LRA, true peak)	EBU Tech 3341 to **0.1 LU**, 3342 to **1 LU**
levels — sample peak/hold, PPM (DIN/Nordic/BBC/EBU), VU	PPM burst to **0.05 dB**; VU 300 ms / 1.5% overshoot
spectrum — FFT analyser, 1/3-octave RTA, spectrogram	pink flat to **1.5 dB**, 25 Hz–16 kHz
transfer function + **coherence** + delay finder	+6 dB peak to **0.3 dB**, coherence > 0.95; sub-sample delay
31-band graphic EQ, master bus	exact at band centres — **12.00 dB for a 12 dB request**; transparent at rest
parametric EQ (per-band Q, frequency, gain)	`Eq::magnitude(hz)` exposes the realised curve

So the measuring instruments are real. What is **not** present yet: a stimulus generator, room acoustics (RT60/EDT/C50/C80/STIPA), modal analysis, a battery orchestrator, the per-venue certificate, and a correction engine that targets *modes* rather than *bands*.

The finding that shapes the whole design

A 1/3-octave graphic EQ is the wrong instrument for correcting a room. A 12 dB cut at 63 Hz at the

graphic's Q of 4.32 spills **9.29 dB into the neighbouring third-octave**. Narrow it and the spill falls (0.68 dB at Q = 10, 0.18 dB at Q = 20) — but a graphic EQ cannot place a narrow notch at a mode.

That single number decides the architecture — and it means the correction **cannot be a set figure**.

Not a preset curve, not 31 fixed bands, not a factory "room" setting. What it has to be:

what	how it is set	why not a fixed figure
how many bands	one per identified mode, plus one shallow band for the broad tilt	the count is whatever the room has; a fixed 31 is 31 guesses
frequency	the measured frequency of that mode	a mode sits where it sits; the nearest third-octave is a different frequency
Q (sharpness)	from the mode's measured width: `Q = f / Δf` at half power	sharpness is a property of the mode's damping, not a taste — a wide mode wants a wide cut, a narrow one wants a narrow one
gain	from the measured excess at that mode, always a **cut**; boosts only for broad dips, never for modes, never beyond +6 dB	a boost at a mode adds energy at the frequency that already rings
verification	against the **realised** response, and against the **worst seat** — never against the requested band values	the filter's actual curve differs from the numbers you asked for; checking the request checks nothing

The parametric band already exists in the tree (`Band::peak(freq, gain, q)`, with `Eq::magnitude(hz)`

reporting what was really applied). The work is not a new filter — it is **computing the set per room**,

then verifying the result as a curve and as a per-seat outcome.

And the limit that keeps it honest: the sharpness is bounded below by what the filter can actually

place. Beyond roughly Q 25 a notch is so narrow in Hz that its own ringing and the realised-versus-requested

difference start to matter, and below about 40 Hz the wavelength is longer than the room's smallest

dimension. Where a mode is narrower than the filter can cut, the report says so — it does not pretend.

What the analysis adds — three questions, and the honesty in each

1. How many microphones do you need? Answered by nudging each unknown in the room model and counting

which directions the measurements actually move.

stage	answer	what it means
what is visible at all	one microphone separates every resonance in the 1-D case	true, and *false to publish alone*
how well separated	one omni leaves **six** directions ten orders below the rest — noise, not information	**two** is the real minimum
at real noise	at **−60 dB**, about **3 of 36** parameters to 10%; at **−50 dB**, none	the honest number

Publishing the first line alone would have been a confident false claim. The useful statement is:

*at normal measurement noise you can reliably measure a handful of combinations, not a table of every

resonance* — which is exactly what an equaliser and an anti-feedback system need, and not what a

"here are your 36 resonances" readout needs.

2. Is the room solver right? A room's resonances have a spacing signature. A plain box produces clustering modes (Poisson); an irregular room produces repelling ones (the same signature the Riemann zeros have). Both are known in advance from the room's shape, so a solver can be checked **without knowing the answer — **and there is also a Weyl count check for the number of modes, measured to** 0.02%** on analytic boxes. Nobody else applies this to room solvers.

3. What should the corrector actually do? Target the modes that were identified, minimise the **worst seat** rather than the average curve, and always report per seat. On a synthetic room the target is a **≥ 6 dB** reduction in worst-seat ripple with no band boosted beyond +6 dB; on a real room, the report names every mode it did **not** fix, and why.

The binaural node — three jobs, and which are measured

The capture hardware is a **binaural node**: two capsules 17 cm apart, stereo in, on batteries, over Wi-Fi. Not a microphone — a sensor with two correlated channels, with three planned jobs:

1. **Walking capture.** The engineer walks and presses; the desk already has the remote. **Measured:** one binaural position does the work of *two* omni positions, so a four-position route can be two — and the system can say "you have enough", and which directions are still weak.
1. **A stationary pair at front of house** — a calibrated, always-on reference channel. *Designed, not measured.*
1. **The stage mics as the measurement array.** Every stage mic is a receiver and the desk knows what it is sending, so **the show itself is the stimulus** — no sweeps, no stopping the gig. *Designed, not measured*, with a named risk: the sends are not a known stimulus unless logged, and parameters change during a show.

The node is **not** "a machine that tunes the room". The engineer stays the sensor; the system is the recorder and auditor. At the end the route *is* the report.

The suite, in phases — each with its own pass mark

phase	produces	pass mark
1 · measurement spine	stimulus + deconvolution to an impulse response, coherence per mic, validity gating	loopback flat **±0.25 dB** 20 Hz–20 kHz, delay to **1 sample**; a capture with a dropout is **refused**, not averaged
2 · room acoustics	RT60/EDT/T20/T30, C50/C80, noise floor, STIPA	RT60 within **5%** of a synthetic decay; C50/C80 within **0.5 dB** of a two-path model
3 · modal analysis + correction	mode identification, Weyl and spacing checks, the notch corrector	**≥ 6 dB** worst-seat ripple reduction, no boost beyond +6 dB, every unfixed mode named
4 · the lab product	one command per venue → a **certificate** (gate, result, evidence, producer, falsifier, remedy) with a manifest hash	a third party can verify the file
5 · the claim	"the world's best audio lab" — only after 1–4, defined by evidence, not by adjective	the evidence page is linked

What it lets us do

- **Certify instead of assert.** Every venue visit produces a file with the falsifier and the remedy in it,

checkable by someone who does not trust us.

- **Compare venues as numbers.** Same fields everywhere, so "this room is worse than that one" stops being

an opinion.

- **Size hardware honestly.** Two capsules or two positions, not five.
- **Make claims that survive a hostile engineer.** Margin and correction, not a modal table.
- **Reuse the method.** The same rank/noise/spacing tests answer the same question for speaker placement,

venue tuning, and any "how much can my sensors know" problem — that is the asset, not the room.

What would make this wrong

- The modal model is **linear and low-frequency** — it does not describe a room above a few hundred Hz, and

it says nothing about loudspeaker nonlinearity.

- It says what is **identifiable**, not what is **audible**. Those are different questions.
- Every number moves with the **noise assumption**, which is why the noise figure is printed beside it.
- The stage-mic array assumes a logged stimulus; if the desk's sends are not recorded, the show is not a

measurement.

How to check us

The 1-D case has an **exact analytic answer** — modes at known pitches, shapes in closed form — so no

solver has to be trusted. The 3-D case is a real 6 × 4.5 × 3 m room with the modes below 100 Hz computed

and cross-checked. The instrument figures above come from the AirMatrix tree's own tests. The reasoning,

including the stage where the headline had to be corrected, is in

identifiability-of-room-modes.md.